

DCN · STUDY GUIDE

Computer Networks and the Internet

Chapter 1 — Computer Networking: A Top-Down Approach (8th Ed., Kurose & Ross)

Covers: What the Internet is · the network edge & access networks · physical media · the network core & switching · delay, loss & throughput · protocol layering & encapsulation · network security · Internet history

Format: Definitions, key rules, worked examples, watch-outs, and intuition boxes — followed by a consolidated quick-reference sheet

Reorganized for exam-ready review

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1.1 What Is the Internet?

1.1.1 A Nuts-and-Bolts Description

Definition Host / End System — any device connected to the Internet. "In Internet jargon, all of these devices are called hosts or end systems."

Definition Packet — when one end system has data to send to another, the sending end system segments the data and adds header bytes to each segment. The resulting packages of information are called packets.

Definition Packet Switch — a network device that takes a packet arriving on one of its incoming links and forwards it onto one of its outgoing links. The two most prominent types: **routers** (network core) and **link-layer switches** (access networks).

Definition Route / Path — the sequence of communication links and packet switches traversed by a packet from sending end system to receiving end system.

Definition ISP (Internet Service Provider) — end systems access the Internet through ISPs (residential, corporate, university, WiFi/cellular, etc). Each ISP is itself a network of packet switches and links. Lower-tier ISPs interconnect through national and international upper-tier ISPs.

Definition TCP and IP — the two most important Internet protocols, together known as **TCP/IP**. IP specifies the format of packets sent and received among routers and end systems.

Definition Internet Standards — developed by the **IETF** (Internet Engineering Task Force). Standard documents are called **RFCs** (Requests for Comments) — nearly 9,000 exist. The **IEEE 802 LAN Standards Committee** specifies Ethernet and WiFi standards.

1.1.2 A Services Description

Definition Distributed Applications — Internet applications involving multiple end systems exchanging data with each other. Applications run on **end systems** — not in packet switches in the network core.

Definition Socket Interface — specifies how a program on one end system asks the Internet infrastructure to deliver data to a specific destination program on another end system. "A set of rules that the sending program must follow so that the Internet can deliver the data to the destination program."

Intuition Think of the socket interface like postal service rules (put the letter in an envelope, write an address, seal it, add a stamp, drop in a mailbox). You must follow the postal service's "interface" for them to deliver your letter. Programs must follow the socket interface for the Internet to deliver their data.

1.1.3 What Is a Protocol?

Definition — Reproduce Exactly Protocol — "A protocol defines the format and the order of messages exchanged between two or more communicating entities, as well as the actions taken on the transmission and/or receipt of a message or other event."

Key Properties of Protocols

- Protocols define the **format** of messages
- Protocols define the **order** of messages
- Protocols define the **actions** taken on transmission/receipt of messages or other events
- All activity in the Internet involving two or more communicating remote entities is governed by a protocol
- It takes **two (or more)** communicating entities running the **same** protocol to accomplish a task

Worked Example — HTTP Web Protocol (Figure 1.2)

1. Client sends TCP connection request to Web server
2. Server returns TCP connection reply
3. Client sends `GET http://www.example.com/` (HTTP GET message)
4. Server returns the requested file

Watch Out If two entities run **different** protocols, they cannot interoperate. Just as human conversations break down when one person doesn't understand the other's language, network protocols fail when entities don't share the same protocol.

1.2 The Network Edge

Definition Network Edge — the computers and other devices connected to the Internet; end systems that sit at the edge of the Internet.

Definition Hosts (= End Systems) — called hosts because they host application programs. Divided into **clients** (desktops, laptops, smartphones) and **servers** (powerful machines that store/distribute Web pages, stream video, relay e-mail).

Definition Access Network — the network that physically connects an end system to the first router (the "edge router") on a path to any other distant end system.

1.2.1 Access Networks

Home Access: DSL

Definition DSL (Digital Subscriber Line) — the customer's telco is also its ISP. The home's DSL modem uses the existing telephone line to exchange data with a **DSLAM** (digital subscriber line access multiplexer) at the telco's central office (CO).

Key Rule — Three Simultaneous Channels (Frequency Division)

- **High-speed downstream channel:** 50 kHz to 1 MHz band
- **Medium-speed upstream channel:** 4 kHz to 50 kHz band
- **Ordinary telephone channel:** 0 to 4 kHz band

DIRECTION	RATES
Downstream	24 Mbps and 52 Mbps (standard); 1 Gbps (newest standard, ITU 2014)
Upstream	3.5 Mbps and 16 Mbps

Because downstream and upstream rates differ, the access is said to be **asymmetric**.

Watch Out DSL has a **distance limitation** — if the residence is not within 5 to 10 miles of the CO, it must use an alternative form of Internet access.

Home Access: Cable

Definition HFC (Hybrid Fiber Coax) — cable Internet access using the cable TV company's existing infrastructure. Fiber optics connect the cable head end to neighborhood-level junctions, from which traditional coaxial cable reaches individual homes — hence "hybrid."

Definition CMTS (Cable Modem Termination System) — at the cable head end; serves a similar function to the DSL DSLAM — turns the analog signal from cable modems back into digital format.

STANDARD	DOWNSTREAM	UPSTREAM
DOCSIS 2.0	40 Mbps	30 Mbps
DOCSIS 3.0	1.2 Gbps	100 Mbps

Watch Out Cable is a **shared broadcast medium**. Every packet sent by the head end travels downstream on every link to every home. If several users simultaneously download a video, each user's actual rate will be significantly lower than the aggregate downstream rate. An upstream distributed multiple-access protocol is needed to avoid collisions.

Home Access: FTTH (Fiber to the Home)

Definition FTTH — provides an optical fiber path from the CO directly to the home; Internet access in the gigabits-per-second range. Two competing architectures:

ARCHITECTURE	DESCRIPTION
AON (Active Optical Networks)	Essentially switched Ethernet
PON (Passive Optical Networks)	Each home has an ONT (Optical Network Terminator) connected by dedicated optical fiber to a neighborhood splitter. The splitter combines homes (typically <100) onto a single shared fiber to an OLT (Optical Line Terminator) at the telco's CO

Enterprise and Home: Ethernet and WiFi

Definition LAN (Local Area Network) — used on corporate/university campuses and increasingly at home to connect end systems to the edge router. Ethernet access rates: 100 Mbps to tens of Gbps (users); 1-10 Gbps (servers).

Definition WiFi (IEEE 802.11) — wireless LAN access. Users must typically be within a few tens of meters of the access point. Shared transmission rate up to more than 100 Mbps.

Wide-Area Wireless: 3G, 4G, 5G

Unlike WiFi, cellular (wide-area) wireless users need only be within a few **tens of kilometers** of the base station. **4G**: real-world download speeds up to 60 Mbps.

1.2.2 Physical Media

Definition Physical Medium — the medium over which bits propagate as electromagnetic waves or optical pulses between a transmitter-

receiver pair.

Two Categories

- **Guided media** — waves guided along a solid medium (fiber-optic cable, twisted-pair copper wire, coaxial cable)
- **Unguided media** — waves propagate in the atmosphere and outer space (wireless LAN, digital satellite)

MEDIUM	NOTES
Twisted-Pair Copper	Least expensive, most commonly used guided medium. Two insulated copper wires ~1mm thick, twisted in a spiral to reduce interference. LANs: 10 Mbps-10 Gbps. Cat 6a: 10 Gbps up to 100 m. Also used for DSL.
Coaxial Cable	Two concentric copper conductors. Used in cable TV; cable modems achieve hundreds of Mbps. Can be used as a guided shared medium.
Fiber Optics	Thin, flexible medium conducting light pulses. Up to tens or hundreds of Gbps. Immune to electromagnetic interference, very low attenuation up to 100 km, very hard to tap. Preferred for long-haul. OC-n standard: link speed = n × 51.8 Mbps (OC-1 through OC-768).
Terrestrial Radio	No physical wire needed; can penetrate walls; supports mobility; potentially long distances. Affected by path loss, shadow fading, multipath fading, interference. Three groups: very short distance (1-2 m), local (10s-100s of m), wide area (tens of km).
Satellite Radio	Two types: geostationary (36,000 km above Earth, 280 ms propagation delay, hundreds of Mbps) and LEO (lower altitude, rotates around Earth, may communicate with each other).

Watch Out The actual cost of physical link material is often minor compared with the **labor cost of installation**, which can be orders of magnitude higher.

1.3 The Network Core

Definition Network Core — the mesh of packet switches and links that interconnects the Internet's end systems.

1.3.1 Packet Switching

Store-and-Forward Transmission

Definition Store-and-Forward Transmission — the packet switch must receive the **entire** packet before it can begin to transmit the first bit of the packet onto the outbound link.

Formula — Store-and-Forward End-to-End Delay (Eq. 1.1) $d_{\text{end-to-end}} = N \cdot (L / R)$
L = packet size (bits), R = transmission rate of each link (bits/sec), N = number of links (N-1 routers between source and destination)

Worked Example Source sends 3 packets, each L bits, over 2 links (1 router), each at rate R.

- Time for first packet to reach destination: **2L/R**
- Time for all three packets to arrive: **4L/R**

Intuition With store-and-forward, the router cannot transmit even the first bit until the whole packet is received — analogous to a truck convoy that must wait at a toll booth until **all** trucks in the convoy arrive before any truck can exit.

Queuing Delays and Packet Loss

Definition Output Buffer (Output Queue) — each packet switch has an output buffer for each attached link, storing packets about to be sent. If the link is busy, arriving packets wait in the buffer — producing queuing delay.

Definition Packet Loss — when a packet arrives to find the buffer completely full, either the arriving packet or an already-queued packet is dropped.

Forwarding Tables and Routing Protocols

Definition IP Address — every end system has an address with a hierarchical structure.

Definition Forwarding Table — each router has a table mapping destination addresses (or portions thereof) to outbound links. On packet arrival, the router examines the destination address and looks up the outbound link.

Definition Routing Protocols — special Internet protocols used to automatically set forwarding tables; may determine the shortest path from each router to each destination.

1.3.2 Circuit Switching

Definition Circuit Switching — resources needed along a path (buffers, link transmission rate) are **reserved** for the duration of the communication session. A **circuit** is a dedicated end-to-end connection. Traditional telephone networks are the classic example.

FEATURE	CIRCUIT SWITCHING	PACKET SWITCHING
Resources	Reserved in advance	On demand
Delay	Guaranteed, predictable	Variable (queuing)
Idle resource waste	Yes (idle periods waste capacity)	No (link used only when sending)
Setup needed	Yes (circuit establishment delay)	No
Example	Telephone network	The Internet

Multiplexing in Circuit-Switched Networks

Two Methods to Implement a Circuit in a Link

- FDM (Frequency-Division Multiplexing)** — the link's frequency spectrum is divided among connections; each gets a dedicated band for the duration. Typical telephone band: 4 kHz.
- TDM (Time-Division Multiplexing)** — time is divided into frames of fixed duration, each divided into fixed time slots; the network dedicates one time slot per frame to each connection.

Formula — TDM Transmission Rate Rate per circuit = frame rate × bits per slot
Example: 8,000 frames/sec × 8 bits/slot = 64 kbps per circuit

Worked Example — Circuit Switching File: 640,000 bits. Network: TDM with 24 slots, link rate 1.536 Mbps. Circuit establishment time: 500 msec.

- Each circuit's rate = 1.536 Mbps / 24 = **64 kbps**
- Time to transmit file = 640,000 / 64,000 = **10 seconds**
- Total time = 10 + 0.5 = **10.5 seconds**

Note: transmission time is independent of the number of links.

Worked Example — Packet Switching Efficiency 1 Mbps shared link; each user: 100 kbps when active, active 10% of the time.

- Circuit switching: supports only **10 users** (1 Mbps / 100 kbps)
- Packet switching with **35 users**: probability that >10 users are simultaneously active ≈ **0.0004** → essentially the same performance while supporting 3× more users

1.3.3 A Network of Networks

Evolution of Internet Structure (Structures 1–5)

- **Structure 1:** All access ISPs connect to a single global transit ISP (customer-provider model)
- **Structure 2:** Multiple competing global transit ISPs, requiring interconnection between them
- **Structure 3:** Multi-tier hierarchy — access ISPs → regional ISPs → tier-1 ISPs (≈ 12 exist, e.g., Level 3, AT&T, Sprint, NTT). No group officially sanctions tier-1 status.
- **Structure 4:** Adds **PoP** (Point of Presence — a group of routers at one location where customer ISPs connect), **multi-homing** (an ISP connecting to two or more provider ISPs for redundancy), and **peering** (two nearby ISPs at the same hierarchy level directly connecting, typically settlement-free) plus **IXP** (Internet Exchange Point — a third-party facility where multiple ISPs can peer together; >600 exist today)
- **Structure 5** (today's Internet): Adds **content-provider networks** (e.g., Google's private TCP/IP network spanning the globe, peering directly with lower-tier ISPs to bypass upper tiers)

Tier-1 ISPs do not pay anyone (top of hierarchy). Lower-tier ISPs are customers of higher-tier ISPs.

Intuition The Internet evolved to its complex structure largely due to **economics and national policy**, not purely performance. Content providers like Google built private networks to reduce payments to upper-tier ISPs and gain greater control over service delivery.

1.4 Delay, Loss, and Throughput in Packet-Switched Networks

1.4.1 Overview of Delay in Packet-Switched Networks

Four Types of Nodal Delay

1. **Processing Delay** (d_{proc}) — time to examine the packet's header, determine where to direct it, and check for bit-level errors. Typically microseconds or less in high-speed routers.
2. **Queuing Delay** (d_{queue}) — time waiting in the output buffer for the link to be free. Varies packet to packet; microseconds to milliseconds.
3. **Transmission Delay** (d_{trans}) — time required to push all of the packet's bits into the link.
4. **Propagation Delay** (d_{prop}) — time for a bit to propagate from the beginning of the link to the next router.

Formulas $d_{\text{trans}} = L / R$ (L = packet length in bits, R = link transmission rate)
 $d_{\text{prop}} = d / s$ (d = distance between routers, s = propagation speed)
 $d_{\text{nodal}} = d_{\text{proc}} + d_{\text{queue}} + d_{\text{trans}} + d_{\text{prop}}$

Propagation speed range: 2×10^8 m/s to 3×10^8 m/s (at or slightly below the speed of light). Propagation delays in wide-area networks: on the order of milliseconds.

Watch Out Transmission delay $\neq$ Propagation delay. Transmission delay is a function of packet length and link rate — nothing to do with distance. Propagation delay is a function of distance — nothing to do with packet length or link rate.

Intuition — Tollbooth Analogy Cars = bits; caravan = packet; tollbooth = router; highway = link. Time for the tollbooth to push the entire caravan onto the highway = transmission delay. Time for one car to travel from one tollbooth to the next = propagation delay.

1.4.2 Queuing Delay and Packet Loss

Formula — Traffic Intensity Traffic Intensity = $\lambda a / R$
a = average rate of packet arrivals (pkts/sec), R = transmission rate (bits/sec), L = packet length (bits)

Key Rules

- If **$\lambda a / R > 1$** : arrival rate exceeds transmission rate $\rightarrow$ queue grows without bound $\rightarrow$ queuing delay approaches infinity
- If **$\lambda a / R \leq 1$** : queuing delay depends on the nature of traffic — periodic arrivals (one packet every L/R sec) give **zero** queuing delay; bursty arrivals give significant average queuing delay
- As traffic intensity approaches 1, average queue length grows **nonlinearly** — a small percentage increase in intensity yields a much larger percentage increase in delay

Golden Rule "Design your system so that the traffic intensity is no greater than 1."

Because queue capacity is finite, when a packet arrives to find a full queue, the router **drops** it — **packet loss**. The fraction of lost packets increases as traffic intensity increases.

Watch Out From an end-system viewpoint, packet loss looks like a packet being transmitted into the network but never emerging at the destination. A lost packet may be retransmitted end-to-end.

1.4.3 End-to-End Delay

Formula — End-to-End Delay (Eq. 1.2, uncongested, homogeneous) $d_{\text{end-end}} = N(d_{\text{proc}} + d_{\text{trans}} + d_{\text{prop}})$
N-1 routers between source/destination, no queuing, processing delay d_{proc} at each node, transmission rate R, propagation delay d_{prop} on each link, $d_{\text{trans}} = L/R$. Generalizes Eq. 1.1, which omitted processing/propagation delays.

Traceroute

Definition Traceroute — a program that sends N special packets toward a destination (one per router on the path). Each router n returns a message to the source on receiving packet n; the destination does the same on receiving packet N. The source records name, address, and round-trip delay to each router. The experiment repeats 3 times (3·N packets total); output includes 3 round-trip measurements per router. An asterisk (*) appears when fewer than three messages are received (due to packet loss).

Worked Example (from textbook) A big jump from Router 7 to Router 8 corresponds to a transatlantic fiber-optic link (large propagation delay).

Watch Out Due to variable queuing delay, the round-trip delay to router n can sometimes be **larger** than the round-trip delay to router n+1. This is not an error.

Definition — Other End-System Delays Media packetization delay (VoIP) — the sender must fill a packet with encoded digitized speech before sending. This can significantly impact user-perceived quality.

1.4.4 Throughput in Computer Networks

Definition Instantaneous Throughput — the rate (bits/sec) at which the destination host is receiving a file at any instant.

Definition Average Throughput — for a file of F bits transferred in T seconds: F/T bits/sec.

Formulas — Bottleneck & Transfer Time Throughput = $\min\{R_s, R_c\}$ (two-link path: server rate R_s , client access rate R_c)
Throughput = $\min\{R_1, R_2, \dots, R_N\}$ (N-link path)
 $T = F / \min\{R_s, R_c\}$ (approximation — ignores store-and-forward, processing delays, and protocol issues)

Bottleneck Link — the link with the minimum transmission rate along a path.

Worked Example MP3 file: $F = 32$ million bits. Server link $R_s = 2$ Mbps. Client access link $R_c = 1$ Mbps. Bottleneck = $R_c = 1$ Mbps.

Transfer time = $32,000,000 / 1,000,000 = \mathbf{32 \text{ seconds}}$

Worked Example — Shared Bottleneck (Multiple Flows) 10 simultaneous downloads sharing a common core link of rate $R = 5$ Mbps. $R_s = 2$ Mbps, $R_c = 1$ Mbps.

- Each download gets $R/10 = \mathbf{500 \text{ kbps}}$ from the common link
- $500 \text{ kbps} < \min\{R_s, R_c\} = 1 \text{ Mbps} \rightarrow$ bottleneck is the **core link**, not the access links
- Each download's throughput = **500 kbps**

Intuition The constraining factor for throughput in today's Internet is typically the **access network**, because the core is generally over-provisioned with high-speed links. However, when many flows share a common link, that shared link can become the bottleneck.

1.5 Protocol Layers and Their Service Models

1.5.1 Layered Architecture

Definition Layered Architecture — an organization of network protocols (and the hardware/software implementing them) into layers. Each layer provides its service by (1) performing certain actions within that layer and (2) using the services of the layer directly below it.

Definition Service Model of a Layer — the services that a layer offers to the layer above.

Advantages of Layering

1. Provides a structured way to discuss complex system components
2. **Modularity** — changing the implementation of a service in one layer doesn't affect other layers, as long as the service interface remains the same

Watch Out — Potential Drawbacks of Layering

1. One layer may duplicate lower-layer functionality (e.g., error recovery at both per-link and end-to-end levels)
2. Functionality at one layer may need information present only in another layer (violates separation of layers)

Definition Protocol Stack — the protocols of the various layers, taken together.

LAYER	NAME	PACKET NAME	PROTOCOL EXAMPLES	IMPLEMENTED IN
5	Application	Message	HTTP, SMTP, FTP, DNS	Software in end systems
4	Transport	Segment	TCP, UDP	Software in end systems
3	Network	Datagram	IP, routing protocols	Mix of hardware & software
2	Link	Frame	Ethernet, WiFi, DOCSIS	Network interface card
1	Physical	Bits	Twisted-pair / fiber protocols	Hardware in link

Layer Descriptions

- **Application Layer** — where network applications and their protocols reside (HTTP, SMTP, FTP, DNS). A packet here is called a **message**.
- **Transport Layer** — transports application-layer messages between application endpoints. Two protocols: **TCP** (connection-oriented, guaranteed delivery, flow control, congestion control, breaks long messages into segments) and **UDP** (connectionless, no reliability/flow/congestion control). A packet here is called a **segment**.
- **Network Layer** — moves datagrams from one host to another. Contains the IP protocol (defines datagram fields and how hosts/routers act on them) and routing protocols. Often called the **IP layer** — there is only one IP protocol.
- **Link Layer** — moves a packet from one node to the next node in the route (not end-to-end). Services depend on the link-layer protocol (Ethernet, WiFi, DOCSIS). Packets here are called **frames**. Some link-layer protocols provide reliable delivery over one link.
- **Physical Layer** — moves individual bits within a frame from one node to the next. Protocols are link-dependent and depend on the transmission medium.

Which Layers Do Different Devices Implement?

- **End systems (hosts):** all 5 layers
- **Routers:** bottom 3 layers (physical, link, network/IP)
- **Link-layer switches:** bottom 2 layers (physical, link)

Intuition "The Internet architecture puts much of its complexity at the edges of the network." Routers and switches only need to process lower layers; the complexity of application and transport protocols lives at end systems.

1.5.2 Encapsulation

Definition Encapsulation — the process by which each layer adds its own header information to the packet received from the layer above, creating a new packet type for that layer.

Encapsulation Process (Sending Side)

1. Application layer creates a message (M)
2. Transport layer takes M, appends transport-layer header (H_t) $\rightarrow$ segment ($H_t + M$)
3. Network layer takes the segment, appends network-layer header (H_n) $\rightarrow$ datagram ($H_n + H_t + M$)
4. Link layer takes the datagram, appends link-layer header (H_l) $\rightarrow$ frame ($H_l + H_n + H_t + M$)

At each layer, a packet has two types of fields: header fields and a payload field. The payload is typically a packet from the layer above. **De-encapsulation** happens in reverse at the receiving end.

Intuition — Interoffice Memo Analogy Memo = application-layer message. Interoffice envelope (with recipient's name/dept) = transport-layer segment. Postal envelope (with full addresses) = network-layer datagram.

Watch Out A large message may be divided into multiple transport-layer segments, each potentially divided into multiple network-layer datagrams. At the receiver, the segment must be **reassembled** from its constituent datagrams.

1.6 Networks Under Attack

1. Malware

Definition Malware — malicious software that enters and infects devices. Can delete files, install **spyware** (collects SSNs, passwords, keystrokes and sends them to attackers), or enroll the device in a **botnet** — a network of thousands of similarly compromised devices controlled by attackers, used for spam distribution or DDoS attacks.

2. Denial-of-Service (DoS) Attacks

Definition DoS Attack — renders a network, host, or other infrastructure unusable by legitimate users.

Three Categories

- **Vulnerability attack** — sending a few well-crafted messages to a vulnerable application/OS — can stop the service or crash the host
- **Bandwidth flooding** — sending a deluge of packets so the target's access link becomes clogged
- **Connection flooding** — establishing many half-open or fully open TCP connections at the target host

Definition DDoS (Distributed DoS) — the attacker controls multiple sources (often a botnet), each sending traffic at the target. Aggregate traffic rate needs to be approximately R (the server's access rate) to cripple the service. Much harder to detect and defend against than single-source DoS.

3. Packet Sniffing

Definition Packet Sniffer — a passive receiver that records a copy of every packet that flies by. Can capture passwords, SSNs, trade secrets, and private messages.

- Difficult to detect because sniffers are **passive** — they do not inject packets into the channel
- Deployed in wireless environments and wired broadcast environments (e.g., Ethernet LANs, cable access networks)
- Best defenses involve **cryptography**

4. IP Spoofing

Definition IP Spoofing — the ability to inject packets into the Internet with a false source address, masquerading as another user.

Definition End-Point Authentication — a mechanism to determine with certainty if a message originates from where we think it does. The solution to IP spoofing.

Watch Out — Why the Internet Is Insecure The Internet was originally designed based on a model of "a group of mutually trusting users attached to a transparent network." Many aspects of the original architecture reflect this — user identity is taken at declared face value, and the ability to send a packet to any user is the default rather than a granted capability.

1.7 History of Computer Networking and the Internet

1.7.1 The Development of Packet Switching: 1961-1972

- **1961:** Leonard Kleinrock (MIT grad student) published the first work on packet switching using queuing theory, demonstrating its effectiveness for bursty traffic
- **1964:** Paul Baran (Rand Institute) investigated packet switching for secure military voice networks; Donald Davies and Roger Scantlebury at NPL (England) developed similar ideas
- **1967:** Lawrence Roberts published the overall plan for ARPANet
- **September 1969:** First packet switch installed at UCLA; four nodes by end of 1969 (UCLA, SRI, UC Santa Barbara, University of Utah)
- **1972:** ARPANet grew to ~15 nodes; first public demonstration by Robert Kahn. First host-to-host protocol (**NCP** — Network Control Protocol) completed. Ray Tomlinson wrote the first e-mail program.

1.7.2 Proprietary Networks and Internetworking: 1972-1980

- New networks: ALOHAnet (Hawaii microwave), DARPA packet-satellite and packet-radio, Telenet (commercial), Cyclades (France, Louis Pouzin), IBM's SNA
- Vinton Cerf and Robert Kahn [Cerf 1974] pioneered internetworking — the term "internetting" was coined
- TCP was developed (initially combining reliable delivery **AND** forwarding functions)
- Recognition that some applications (like packetized voice) needed unreliable transport → IP separated from TCP → UDP developed
- TCP, UDP, and IP were conceptually in place by the end of the 1970s
- **ALOHAnet** (Norman Abramson): first multiple-access protocol for a shared broadcast medium
- **Ethernet** (Metcalfe and Boggs): built on ALOHA's multiple-access work for wire-based shared broadcast networks

1.7.3 A Proliferation of Networks: 1980-1990

- By end of 1970s: ~200 hosts on ARPANet; by end of 1980s: ~100,000 hosts
- New networks: BITNET, CSNET, NSFNET (1986, started at 56 kbps backbone, reached 1.5 Mbps by end of decade)
- **January 1, 1983:** official deployment of TCP/IP as the new standard host protocol for ARPANet (replacing NCP) — a "flag day" event
- DNS developed [RFC 1034] to map human-readable names to 32-bit IP addresses
- Important TCP extensions for host-based congestion control [Jacobson 1988]
- France's Minitel project: public packet-switched network (X.25 protocol), free terminals to households. Peak: 20,000+ services.

1.7.4 The Internet Explosion: The 1990s

- ARPANet ceased to exist
- **1991:** NSFNET lifted restrictions on commercial use
- **1995:** NSFNET decommissioned; Internet backbone carried by commercial ISPs
- World Wide Web invented by **Tim Berners-Lee** at CERN between 1989-1991; developed HTML, HTTP, a Web server, and a browser
- ~200 Web servers by end of 1993; browser GUI by Marc Andreessen → Netscape (1994-1995)
- **1996:** Microsoft entered the browser market → browser war (Microsoft won)

Four Killer Applications of the Late 1990s

1. E-mail (including attachments, Web-accessible)
2. The Web (browsing and e-commerce)
3. Instant messaging (with contact lists)
4. Peer-to-peer MP3 file sharing (Napster)

Internet stocks collapsed 2000-2001; winners included Microsoft, Cisco, Yahoo, eBay, Google, Amazon.

1.7.5 The New Millennium

Key Developments

- Aggressive deployment of broadband (cable, DSL, fiber, 5G fixed wireless)
- High-speed wireless Internet (smartphone revolution) — wireless devices surpassed wired in 2011
- Online social networks (Facebook, Instagram, Twitter, WeChat)
- Content providers building private networks (Google, Microsoft) and peering directly with lower-tier ISPs
- Cloud computing (Amazon EC2, Microsoft Azure, Alibaba Cloud)

★ Quick Reference Sheet

All Formulas

FORMULA	DESCRIPTION
$d_{trans} = L/R$	Transmission delay — L = packet size (bits), R = link rate (bits/sec)
$d_{prop} = d/s$	Propagation delay — d = distance (m), s = propagation speed (m/s)
$d_{nodal} = d_{proc} + d_{queue} + d_{trans} + d_{prop}$	Total nodal delay
$d_{end-end} = N \cdot (L/R) \quad (\text{Eq. 1.1})$	End-to-end delay, N links, no proc/prop/queue
$d_{end-end} = N(d_{proc} + d_{trans} + d_{prop}) \quad (\text{Eq. 1.2})$	End-to-end delay including proc & prop, no queuing — N = number of hops, homogeneous links
Traffic Intensity = $\lambda a/R$	Key queuing metric — L = packet size, a = arrival rate (pkts/sec), R = link rate
Throughput = $\min\{R_1, \dots, R_N\}$	Bottleneck throughput for an N-link path
$T_{file} = F / \min\{R_s, R_c\}$	File transfer time (approx.) — F = file size (bits), R_s = server link rate, R_c = client link rate
Rate = frame rate × bits/slot	TDM circuit rate
Bit width = s/R	Physical width (meters) of one bit on a link — s = propagation speed, R = transmission rate

All Key Definitions

TERM	DEFINITION
Host / End System	Any device connected to the Internet
Packet	A chunk of data with header bytes, created by segmenting a larger message
Packet Switch	Forwards packets from incoming links to outgoing links; routers and link-layer switches
Route / Path	The sequence of links and packet switches a packet traverses source → destination
ISP	Internet Service Provider; networks through which end systems access the Internet
TCP/IP	The principal protocols of the Internet (Transmission Control Protocol / Internet Protocol)
RFC	Request for Comments; IETF standards documents defining Internet protocols
Protocol	Defines format, order of messages, and actions taken on transmission/receipt between entities
Distributed Application	Application involving multiple end systems exchanging data with each other
Socket Interface	Rules a sending program must follow so the Internet delivers data to the destination program
Access Network	Network physically connecting an end system to the first (edge) router
DSL	Digital Subscriber Line; uses existing telephone lines; asymmetric; DSLAM at CO
DSLAM	Digital Subscriber Line Access Multiplexer; at telco's CO, separates data and phone signals
HFC	Hybrid Fiber Coax; cable Internet using fiber to neighborhood junctions, then coaxial to homes
CMTS	Cable Modem Termination System; at cable head end, converts analog to digital
FTTH	Fiber to the Home; optical fiber from CO to home; gigabit-range speeds
PON	Passive Optical Network; FTTH architecture — ONT at home, splitter combining homes onto shared fiber to OLT
WiFi (IEEE 802.11)	Wireless LAN standard; up to ~100 Mbps+; users within tens of meters of access point
Guided Media	Media where waves are guided along a solid medium (fiber, copper wire, coaxial)
Unguided Media	Media where waves propagate in atmosphere/space (wireless)
Store-and-Forward	Packet switch must receive entire packet before transmitting onto outbound link
Output Buffer	Queue at a packet switch where packets wait when the link is busy
Packet Loss	Packet dropped when the output buffer is full
Forwarding Table	Router's table mapping destination addresses to outbound links
Routing Protocol	Protocol that automatically sets forwarding tables (e.g., finds shortest paths)
Circuit Switching	Resources reserved end-to-end for the duration of a session (e.g., telephone network)
FDM	Frequency-Division Multiplexing; allocates a frequency band to each circuit for the connection's duration
TDM	Time-Division Multiplexing; allocates time slots in revolving frames to each circuit

Traffic Intensity	λ/R ; ratio of bit arrival rate to link capacity; must be ≤ 1 to avoid infinite queuing delay
Bottleneck Link	Link with minimum transmission rate along a path; determines end-to-end throughput
Instantaneous Throughput	Rate (bits/sec) at which the destination is receiving a file at any instant
Average Throughput	F/T for a file of F bits transferred in T seconds
Layered Architecture	Protocol organization into layers where each layer uses the services of the layer below
Protocol Stack	All protocols across all layers, taken together
Message	Packet name at the application layer
Segment	Packet name at the transport layer
Datagram	Packet name at the network layer
Frame	Packet name at the link layer
Encapsulation	Process by which each layer adds header information to the packet from the layer above
Payload	The layer-above packet carried inside the current layer's packet
Malware	Malicious software that infects devices (spyware, viruses, etc.)
Botnet	Network of compromised hosts controlled by attackers
DoS Attack	Renders infrastructure unusable; types: vulnerability, bandwidth flooding, connection flooding
DDoS Attack	DoS attack from many controlled sources simultaneously
Packet Sniffer	Passive receiver recording copies of every packet passing by
IP Spoofing	Injecting packets with false source addresses
End-Point Authentication	Mechanism to verify the true origin of a message
Tier-1 ISP	Top-level ISPs (≈ 12 exist) that don't pay anyone; connected to each other
IXP	Internet Exchange Point; third-party facility where multiple ISPs peer together
Peering	Two ISPs at the same hierarchy level directly connecting, typically settlement-free
Multi-homing	An ISP connecting to two or more provider ISPs for redundancy
PoP	Point of Presence; group of routers at one location where customer ISPs connect to a provider
Traceroute	Program sending N special packets to measure round-trip delay to each router along a path
NCP	Network Control Protocol; first host-to-host ARPAnet protocol (1972), replaced by TCP/IP in 1983
ARPAnet	First packet-switched computer network (1969); direct ancestor of today's Internet

Example Problem Types Covered

- Store-and-forward delay — total delay for sending P packets over N links at rate R (Eq. 1.1)
- End-to-end delay — total delay including propagation and processing (Eq. 1.2)
- Transmission delay vs. propagation delay — distinguish and compute each ($d_{\text{trans}}=L/R$; $d_{\text{prop}}=d/s$)
- Circuit switching — per-circuit rate (TDM), file transfer time including circuit setup
- Packet switching vs. circuit switching efficiency — max users under circuit switching; probability calculations for packet switching supporting more users
- Traffic intensity — compute λ/R ; determine whether queuing delay is acceptable
- Throughput — identify bottleneck link; compute file transfer time $F/\min\{R_1, \dots, R_N\}$
- Throughput with shared core link — divide shared link R among M flows to find per-flow throughput
- Queuing delay formula — use $\frac{\lambda L}{R(1-I)}$ for queuing delay given traffic intensity I (Homework P14)
- VoIP packetization delay — time for a bit to be created at sender and decoded at receiver
- Traceroute analysis — interpret router names, addresses, round-trip delays; identify transatlantic link jumps
- Bandwidth-delay product — compute $R \times d_{\text{prop}}$; interpret as bits "in flight" in the link
- Message segmentation — compare transfer time with and without segmentation over N links
- Little's formula — $N = a \times d$ to find packet arrival rate given buffer size and delay
- General end-to-end delay — heterogeneous links with different rates, distances, and processing delays (Homework P10, P17)